XGBoost ML Algorithm (classification)

Data Set

Salary	Credit	Approval
<=50K	B	0
<=50K	G	1
<=50K	G	1
>50K	B	0
>50K	G	1
>50K	N	1
<50K	N	0

Steps

1. construct a base model
2. construct a Decision Tree with root.
3. Calculate similarity weight

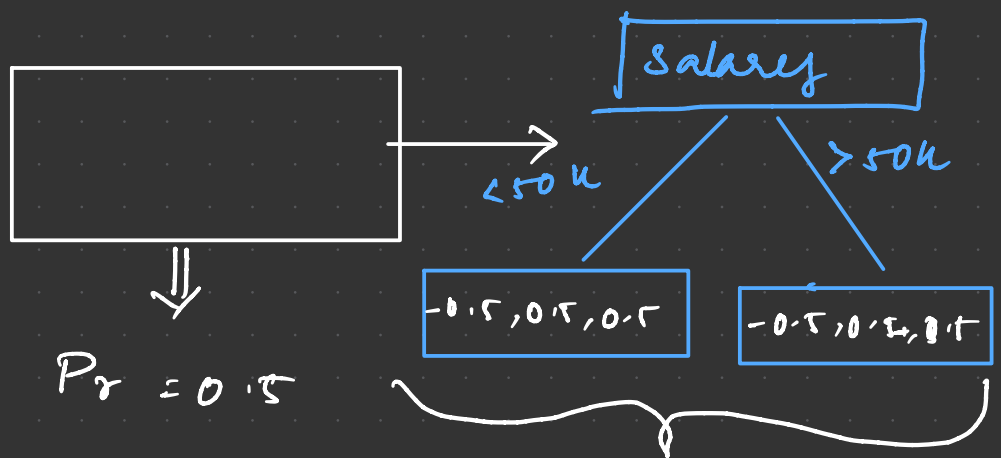
$$= \frac{\Sigma (\text{Residual})^2}{\Sigma Pr(1-Pr) + \lambda}$$

4. Calculate Gain

$\lambda \rightarrow$ hyperparameter

Base model

1



These are nothing but the residuals, we calculate residuals using $R_1 = \hat{y} - y_i$ → here $\hat{y} = 0.5$ as the probability is 0.5

Similarity weight for the left child

$$\text{Similarity weight}(LC) = \frac{\sum (\text{Residual})^2}{\sum P_r (1 - P_r)}$$

$$= \frac{\left[(-0.5 + 0.5 + 0.5 - 0.5)^2\right]}{\left[0.5(-0.5) + 0.5(0.5) + 0.5(0.5) + 0.5(-0.5)\right]}$$

$$= 0$$

Similarly,

Similarity weight for the right child

$$\text{Similarity weight}(RC) = 0.33$$

Similarity weight for the root node

$$\text{Similarity weight}(\text{Root}) = 0.14$$

Compute Gain

$$\text{Gain} = (LC + RC) - \text{Root}$$

$$= (0 + 0.33) - 0.14$$

$$= 0.21$$

Similarly, we will have to repeat the same steps with the other feature. in our case it will be credit.

Finally, we will be selecting the node which has the maximum Gain.

For any new test data.

we will compute

$$\log(\text{odds}) = \log\left(\frac{P_r}{1 - P_r}\right)$$

$$= \log\left(\frac{0.5}{0.5}\right)$$

$$= 0$$

then, apply sigmoid activation

$$\xrightarrow[\text{data}]{\text{new}} = \sigma\left(\log(\text{odds}) + \alpha(\cdot)\right)$$

$$\boxed{= \frac{1}{1 + e^{-\lambda(z)}}}$$

$\Rightarrow$ This will be computed for any new test data.

The output of the sigmoid function will be considered / treated as $\hat{y}$

Sigmoid function is also called as Logistic function.